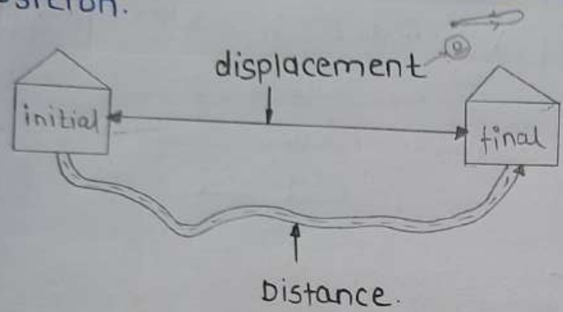
MOTION

Displacement :

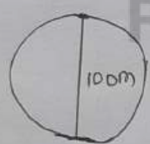
Shortest distance measured from the initial to the Final position.



PYQ 2021

1] An athlete one round of a circular track of diameter 100m in 20sec. What will be the displacement after 1minute & 10s. respectively.

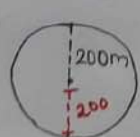
→ every 20 sec. displ. is 0.



$\boxed{0m \; \& \; 100m}$

2] Circular track, radius = 200m
1 revolution 20 sec.
after 1:30min displacement = ?

→ 1 rev. – 20 sec.
Total t. – 90 sec



80sec – 0m
+
10 sec = 200×2
= 400 m

distance travelled – odometer

uniform motion (speed constant)	non-uniform
- equal distance in equal interval of time.	- unequal distance in equal interval of time.
- velocity constant	- velocity can't constant.

Avg. speed = $\frac{\text{Total distance travell.}}{\text{total time taken}}$

Velocity = $\frac{\text{displacement}}{\text{time}}$

Acceleration (a) = $\frac{\text{change in velocity}}{\text{time taken}}$

- m/s $\xrightarrow{18/5}$ km/h
- km/h $\xrightarrow{5/18}$ m/s

magnitude & velocity – Vector

speed – scalar quantity.

$\boxed{\text{Deacceleration} = -a}$

uniform Acceleration :

→ velocity increase or decreases by equal Amount in equal interval of time.

velocity :

speed of an object moving in a definite direction.

Formula

(s - distance)

1) $v = u + at$

2) $s = ut + \frac{1}{2}at^2$ (distance?)

3) $2as = v^2 - u^2$

1] A dog running at a speed of 72 km/h slowed down to 18 km/h over a distance of 40m. calculate the retardation.

→ $u = 72$ km/h $= \cancel{72}^{4} \times \frac{5}{\cancel{18}} = 20$ m/s

$v = 18$ km/h $= 18 \times \frac{5}{\cancel{18}} = 5$ m/s

$d = 40$m

$v^2 - u^2 = 2as$

$400 - 25 = 2 \times a \times 40$

$-\frac{375}{80} = a$

$\boxed{a = -4.68 \text{ m/s}^2}$

2] A train Acc. from 36 km/h to 54 km/h in 10 sec. Find acc. & distance.

→ $u = 36$ km/h $= \cancel{36}^{2} \times \frac{5}{\cancel{18}} = 10$ m/s

$v = 54$ km/h $= \cancel{54}^{3} \times \frac{5}{\cancel{18}} = 15$ m/s

$t = 10$ sec

$\therefore \boxed{v = u + at}$

$a = \frac{v - u}{t}$

$= \frac{15 - 10}{10} = \frac{5}{10} = 0.5$ m/s

$\boxed{s = ut + \frac{1}{2}at^2}$

$s = 10 \times 10 + \frac{1}{2} \times 0.5 \times (10)^2$

$= 100 \times \frac{1}{\cancel{2}} \times 0.5 \times \cancel{100}\,50$

$= 100 + 25$

$= 125$ m

3] A bullcart at rest Accelerated uniformaly to a speed of 144 km/h in 20 sec then distance?

→ $V = \cancel{144} \times \frac{5}{\cancel{18}} = 40$ m/s

$u = 0$

$a = \frac{v - u}{t}$

$a = \frac{40 - 0}{\cancel{20}}$

$a = 2$

$s = ut + \frac{1}{2}at^2$

$= 0 + \frac{1}{\cancel{2}} \times 2 \times \cancel{400}\,200$

$= 400$ m

$= 0.4$ km

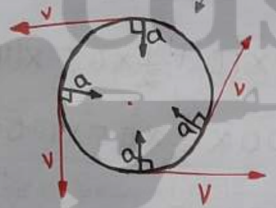
* the direction of Acceleration in uniform circular motion

↓

perpendicular with velocity inwards.

* Uniform circular motion

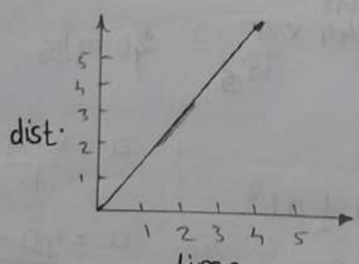
- Speed is constant
- velocity is not constant.
- velocity - tangent to the circle.



Accel. - Inwards

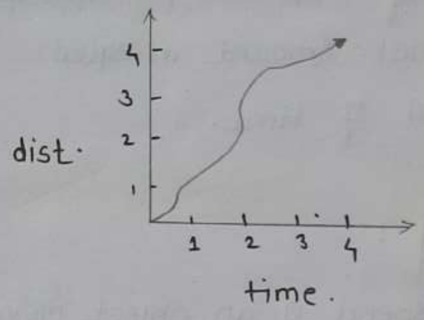
* Graphical representation

* Uniform motion:



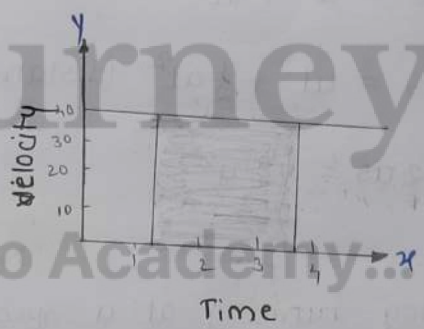
→ for uniform speed : a graph of distance travelled against time is a straight line.

* Non-uniform motion



- unequal distance in equal interval of time.

* velocity time graph



- If the obj. moves at uniform velocity the height of its velocity-time graph will not change with time.
- It will be a straigh line parallel to the x-axis

* for uniform acceleration:

- Velocity - time graph is a Straight line.

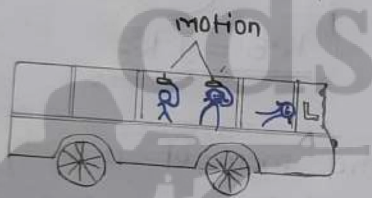
● area under the velocity - time graph gives distance (magnitude of displacement).

non-uniform Acceleration

- velocity - time graph can have any shape.

1st law of motion

- also called law of Inertia.
- object wove with a constant speed unless external force.
- relation beth velocity & time.



Inertia

- measured ⟶ mass
- more mass = more inertia.

1st law of motion :

- when an unbalenced external force Acts on object its velocity changes.

2nd law of motion

- rate of change of momentum of an object is proportional to the applied unbalanced force in the direction of force.
- relation b/w position & time.

(kg m/s)

* Momentum (P) = mv

both

direction magnitude.

velocity also both

*
$$a = \frac{\text{change in velocity}}{\text{time}} = \frac{v-u}{t}$$

$F = ma$ (F : chang in mom.)

$$F = m\frac{(v-u)}{t}$$

$$F = \frac{mv - mu}{t}$$

$$\boxed{F \propto a}$$

- Force = $kg\,ms^{-2}$ or Newton

Third law of motion

- to every action there is an equal and opposite reaction.
- two forces are always equal in magnitude but opposite direction.

$K \cdot E = \frac{1}{2} mv^2$

$P \cdot E = mgh$

$P = mv$ second law

$F = ma$

$K \cdot E = \frac{1}{2} mv^2$ ⊛ mass ↑ - K·E ↓

$\boxed{K \cdot E = \frac{1}{2} \frac{P^2}{m}} = \frac{P^2}{2m}$

$\boxed{P = \sqrt{2m \times K \cdot E}}$

Q. If the K·E of a body becomes 4 times then what will be the change in momentum?

→ $K \cdot E = \frac{P^2}{2m}$

$\boxed{P' = \sqrt{2m \times K \cdot E}}$

$P' = \sqrt{2m \times 4 K \cdot E}$

$P' = 2\sqrt{2mK \cdot E}$

$\boxed{P' = 2P' \text{ time Inc.}}$

Q. If the mom. of a body is tripled its K·E?

① same

② 1/9th

③ 3 times

④ 9 times. ✓

$K \cdot E = \frac{P^2}{2m}$

$K \cdot E = \frac{(3P)^2}{2m}$

$K \cdot E = 9\frac{P^2}{2m}$

$K \cdot E = 9 \; K \cdot E$

Q. If the mom of a body get doubled its K·E will?

① incre. by 2 time

② ↑ 4 time ✓

③ half

④ becomes $\frac{P^2}{m}$ times of old K·E.

$K \cdot E = \frac{P^2}{2m}$

$K \cdot E = \frac{(2P)^2}{2m}$

$K \cdot E = 4\frac{P^2}{2m}$

$K \cdot E = 4 \; K \cdot E$

Q. If the momentum of a body is Numer. equal to its K.E, then the velocity of the body is?

① $1 ms^{-1}$

② $2 ms^{-2}$

③ $2 km h^{-1}$

④ None

$$\frac{1}{2}\cancel{m}v^{\cancel{2}} = \cancel{m}\cancel{v}$$

$$\frac{1}{2} \times v$$

- $\underline{V = 2 ms^{-1}}$

- velocity $= ms^{-1}$
- Acc. $= ms^{-2}$

Q. K.E of a body is Num. equal to trice the momentum. the velocity of the body?

① 2

② 4

③ 6

④ 2/3

$K.E = 3 \times mom.$

$K.E = \frac{1}{2}mv^2$

$\frac{1}{2}\cancel{m}v^{\cancel{2}} = 3 \times \cancel{m} \times \cancel{v}$

$\frac{v}{2} = 3$

- $\underline{\text{velocity} = 6}$

Q. If the velocity is doubled then

→ mom. ↑ by <u>2 times</u>, <u>K.E by 4 time</u>

(because) - $K.E = \frac{1}{2}mv^2$

$= \frac{1}{2}m(2v^2)$

$= \frac{1}{2}m4v^2$

$KE = 4\frac{mv^2}{2}$

$\boxed{K.E = 4}$ — ①

$P = \sqrt{2m \times K.E}$ | $P = mv$

$P = \sqrt{2m \times 4K.E}$ | $= m \times 2 \times v$

$P = 2\sqrt{2m K.E}$ | $P = 2mv$

$\underline{P = 2P'}$ | $\underline{P = 2P'}$

Q. If both mass and velocity of a ball is boubled the K.E becomes

① 4

② 8

③ 16

④ No change

$K.E = \frac{1}{2}mv^2$

$= \frac{1}{2} \times 2m \times 4v^2$

$K.E = 8\frac{mv^2}{2}$

$\underline{K.E = 8}$

$\boxed{\text{Energy} = \text{power} \times \text{time taken}}$

- 1 Watt = 1 J/S
- 1 Joule = 1 Nm

1 KW = 1000 watts

1 KW = 1000 J/s

$\boxed{1 HP = 746 W}$

- $\boxed{\text{electricity common unit} = KWH}$

- $\boxed{1 KWh = 3.6 \times 10^6 J}$

$\boxed{\text{Work}}$ done = Force × displacement

$W = FS$

- Work has only magnitude
- No Direction.

Kinetic energy	potential ener.
• K·E is the energy possessed by an obj. due to its motion.	• The energy transferred to an object is stored as potensial energy
• K·E ↑ − speed ↑ (v)	• $P.E = mgh$
• $K.E = \frac{1}{2}mv^2$	• P·E of an obj. ↑ with its Height

$\boxed{mgh + \frac{1}{2}mv^2 = \text{constant}}$

◉ sum ⇒ K·E + PE = mechanical energy.

$\boxed{\text{density} = \frac{\text{mass}}{\text{Volume}}}$

- velocity ↓ − density ↑
- velocity ↑ − density ↓
- density ↑ − any object sink

during the free fall of obj.

K·E = Increase.

P·E = decrease.

Q. An electric bulb of 60W is used for 6h per day. calc. the 'units' of energy consumed in one day by the bulb.

① 36 unit
② 360 unit
③ 0·36 unit
④ 3·6 unit

Energy = power × time

∴ 60W

$= \frac{60}{1000}$

$P = 0.06$ KW

$t = 6h$

◉ one day

$E = 0.06 \times 6$

$= 0.36$ unit

Q. Fridge of 100W → 2hr in a day What is the cost of using it for 28 days? [1 unit = 3 Rupee].

① 0·6
② 168
③ 600
④ 188

$100W = \frac{100}{1000}$ KW

$= 0.1$ KW

$E = P \times t$

$= 0.1 \times 2$

$\boxed{1 \text{ day} = 0.2}$

28 days → 0.2×28

$= 5.6$ Unit

1 unit = 3 rupee

$= 5.6 \times 3$

$\boxed{= 16.8}$

Q. Find the energy possessed by an object of mass 10 kg when it is at a height of 6m above the ground. ($g = 9.8$)

→

$m = 10\,kg$

$h = 6\,m$

$g = 9.8$

potential energy $= mgh$

$= 10 \times 9.8 \times 6$

$= \underline{\underline{588\,J}}$

* A porter lifts a lauggae of 15kg from the ground and puts it on his head 1.5 m above the ground calculate the work done by on the lauggae.

→

$m = 15\,kg$

$s = 1.5\,m$ (disp.)

Work (W) $= F \times S$

$= mg \times S$

$= 15 \times 10 \times 1.5$

$= 225\,kg\,ms^{-2}\,m$

$= \underline{\underline{225\,J}}$

$\boxed{1\,KJ = 1000\,J}$

Law of conservation of energy

↓

energy can only be converted from one from to another

↓

It can Neither be created or destroyed.

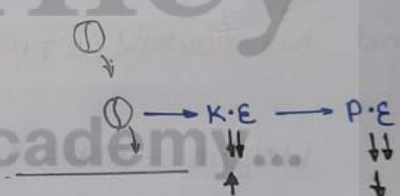
(i)

K.E ← ○ → P.E = mgh.

↓

zero because velocity zero.

(ii)



fall of the object continues the P.E would decrease while

+

K.E would increase.

(iii) When the object is about to reach the ground.

↓

$h = 0$

$v =$ will be higest

* P.E + K.E = constant

↑

menchanical energy

Power

$$\text{power} = \frac{\text{work}}{\text{time}}$$

$$P = \frac{W}{t}$$

- $1\ kW = 1000\ W$
- $1\ kW = 1000\ J s^{-1}$
- $1\ kWh = 3.6 \times 10^6\ J$
- $1\ kWh = 3600000\ J$

Q. A boy of mass 50 kg runs up a staircase of 45 steps in 9s. If the hight of each step is 15cm find his power? ($g = 10\ m/s$)

→

weight = mass x gravity

$W = mg$

$W = 50 \times 10$

$W = 500\ N$

Hight of the staircase

$h = 45 \times \frac{15}{100}\ m$

$h = 6.75\ m.$

$P = \frac{\text{Work done}}{\text{time take}} = \frac{mgh}{t}$

$P = \frac{500\ N \times 6.75\ m}{9\ s}$

$\boxed{P = 375\ W}$

Q. find the energy consumed in 10hr by 4 devices of power 500W each?

→ Energy = power × t

$P = \frac{500}{1000} = 0.5\ kW$

$\boxed{P = 0.5\ kW}$

$\boxed{t = 10\ hr}$

$E = 0.5 \times 10$

$E = 5$ unite

for one device.

$E = 5 \times 4$

$E = 20$ units

Q. find the energy consumed by 10 bulb of 60W each in 5 hours?

① 30 KWH
② 3000KWH
③ 3WH
④ 3KWH

$P = \frac{6\cancel{0}}{100\cancel{0}} = 0.06$

$P = 0.06 \times 5$

$P = 0.30\ kWh$

$P = 0.30 \times 10$

$P = 3\ kWh$

Q. A 100W bulb is used for 5 hr daily and four 50W bulb for 5hr daily calculate the energy consumed in the month of Jan.

→

$$P_1 = \frac{10\cancel{0}}{100\cancel{0}} = 0.1$$

$$E_1 = 0.1 \times 5$$

$$E_1 = 0.5 \quad \text{— (i)}$$

$$P_2 = \frac{5\cancel{0}}{100\cancel{0}} = 0.05$$

$$E_2 = 0.05 \times 5$$

$$E_2 = 0.25$$

$$E_2 = 0.25 \times 4$$

$$= 1 \text{ KWH} \quad \text{— (ii)}$$

$$\text{Total energy} = E_1 + E_2$$

$$= 0.5 + 1$$

$$E = 1.5 \text{ KWH}$$

$$= 1.5 \times 31 \text{ (Jan)}$$

$$= 46.5 \text{ KWH}$$

1HP = 746W

Gravitation

* **centripetal force :**

→ body moving along the circular path is acting towards the centere.

↓

centripetal force : moon more around the earth.

↓

provided by force of Attraction of the earth.

* **Gravitational force**

$$F \propto \frac{1}{d^2}$$

$$F \propto \frac{M_1 \times m_2}{d^2}$$

$$\boxed{F = G\frac{M_1 \times m_2}{d^2}}$$

$$\boxed{G = \frac{F \times d^2}{m_1 \times m_2}}$$

↑

$6.7 \times 10^{-11}\ Nm^2 kg^{-2}$

- Lactometers - determine the purity of sample of milk.
- Hydrometers - determine density of liquids

Q. If the distance b/w two object is doubled the gravit. force b/w them?

① became $\frac{1}{16}$th

② became 4 time

③ gets doubled

④ becames $\frac{1}{4}$th

$$F = \frac{G m_1 \times m_2}{(2d)^2}$$

$$F = G\frac{m_1 \times m_2}{4d^2}$$

$$F = \frac{1}{4} G \frac{m_1 \times m_2}{d^2}$$

$$\boxed{F = \frac{1}{4}F}$$

Q. What happens to the force b/w 2 object If the mass of one of the body is doubled and dist. get halved.

→

① dec. by 16 times

② dec. by 8 times

③ inc by 8 times

④ inc by 16 times.

$$F = \frac{G m_1 \times m_2}{d^2}$$

$$= \frac{G 2 m_1 \times m_2}{(\frac{d}{2})^2}$$

$$= \frac{G \times 2 \times m_1 \times m_2}{\frac{d^2}{4}}$$

$$F = 8\frac{G m_1 \times m_2}{d^2}$$

$$\boxed{F = 8F}$$

universal law of gravitation successfully explained.

i) the force that binds us to the earth.

ii) the motion of the moon around the earth.

iii) the motion of planets around the sun

iv) the tides due to the moon & the sun.

* $W = mg$
* $F = ma$
* $g = G\frac{M}{R^2}$ — M: mass of the earth, R: Radius of earth

→ g becomes greater at poles than equator

→ weight (N): varies from place to place.
force of Attraction of the earth on an obj.

→ moon = $\frac{1}{6}$ x weight of the earth

→ Earth - 5.98×10^{24} (mass)
moon - 7.36×10^{22} (mass)

- magnitude of this buoyant force depends on the density of the fluid.

Q. mass of an object is 10kg what is its weight on the earth.

→
$m = 10\ kg$
$W = ?$
$g = 9.8$

$W = m \times g$
$= 10 \times 9.8$
$W = 98N$

Q. object weight 10N when measured on the surface of the earth. What would be its weight when measured on the surface of the moon.

→

moon = $\frac{1}{6}$ x weight on eart
$= \frac{1}{6} \times 10$
$= \frac{10}{6}$
$= 1.67N$

the force Acting on an object perpenduculer to the surface is called thrust.

$$\text{pressure} = \frac{\text{trust}}{\text{area}}$$

Buoyancy: upward force exerted by the water on the bottle is known as upthrust or buoyant force.

Q. What happens to the weight of a person when lift goes down with Acceleration.

→ Weight of the man becomes <u>zero</u>

Q. What will happen to a person's weight when he is in a moving elevator at a constant speed.

→ weight will not change.

* Archimedes principle

↓

used

↓

designing ships & submarines.

* $\boxed{g = a}$

↓ $g = +ve$, $u = 0$ | ↑ $g(-ve)$, $v = 0$

Q. A ball is thrown vertically upwards with a velocity of 49 m/s calculate.

i) the max. height to which it rises.

ii) the total time it takes to return to the surface of the hight.

→

$v = 0$

$u = 49$

$g = -9.8$ (thrown hight)

i) $v^2 - u^2 = 2as$

$0^2 - 49^2 = 2 \times (-9.8) \times h$

$49^2 = -2 \times 9.8 \times h$

$\boxed{h = 122.5\ m}$ — ①

ii) time

ⓐ return

$v = u + gt$

$0 = 49 - 9.8 \times t$

ऊपर से अगर छोड़ रहे है तो initial velocity छोड़ने वाले point पर 0 होगी

$9.8t = 49$

$\boxed{t = 5s}$ ← जानेका

total time $= 5 \times 2 = 10s$

Q. A stone is thrown vertically upward with an initial velocity of 40 m/s. (Taking $g = 10$)

① find the max. height reached by the stone.

ball ऊपर जा कर रुक रही है.

→ $v = 0$ = (Final velocity)

$u = 40$

$g = 10\ m/s^2$

$h = ?$

$v^2 - u^2 = 2as$

$0 - 1600 = -2 \times 10 \times h$

$= -20 \times h$

$h = \frac{1600}{20} = \boxed{h = 80m}$

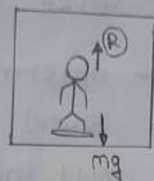
Total distance 160 m

* Apparent weight in lift.

$\boxed{F = mg}$ (#) R - Normal force $\boxed{F = ma}$

i) **Lift at rest (constant velocity)**

$$\boxed{R = mg}$$



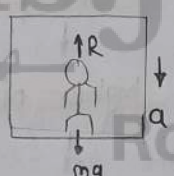
ii) **lift Acceleration upward**

$$R - ma = mg$$
$$R = mg + ma$$
$$\boxed{R = m(g+a)}$$

(+a) ↑ Acc.

iii) **lift Acceleration downward.**

$$R = mg - ma$$
$$\boxed{R = m(g-a)}$$



iv) **during free fall** (7th case)

$$acc.(a) = g$$
$$R = mg + mg$$
$$R = m(g-g)$$
$$\boxed{R = 0}$$

*Acc. is const. throughout the free fall.

Q. The app. weight of a person in a lift moving downwards with Acce. is half his app. weight moving upward in the same lift with same Acc.?

→ (a) g
(b) g/2
(c) g/4
(d) g/3

$$\cancel{m}(g-a) = \frac{1}{2}\cancel{m}(g+a)$$
$$2(g-a) = g+a$$
$$2g - 2a = g + a$$
$$2g = g + 3a$$
$$2g - g = 3a$$
$$g = 3a$$
$$a = \frac{g}{3}$$

Q. A person of mass 60 kg is in lift calculate change in Apparent weight of a person. left Acc. with $2\,m/s^2$ & then down with of $2\,m/s^2$

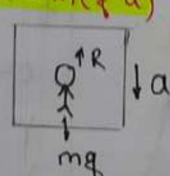
i) $w = m(g+a)$
$w = 60(10+2)$
$\boxed{W = 720N}$

ii) $W = m(g-a)$
$= 60(10-2)$
$= 60 \times 8$
$w\ \boxed{= 480N}$

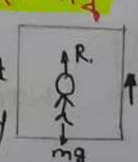
→ $720 - 480 = 240N$

① $R = m(g+a)$ moving up

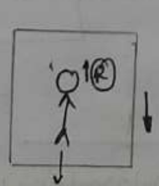
② $R = m(g-a)$ Moving down



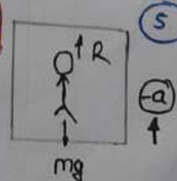
③ $R = mg$ constant velocity



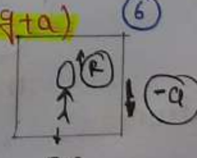
④ constant velocity $\boxed{R = mg}$



⑤ $\boxed{R = m(g-a)}$ Deacc.



⑥ $R = m(g+a)$ Retardation



Sound

Q. Which of the following is not present in vaccum?

ⓐ Light

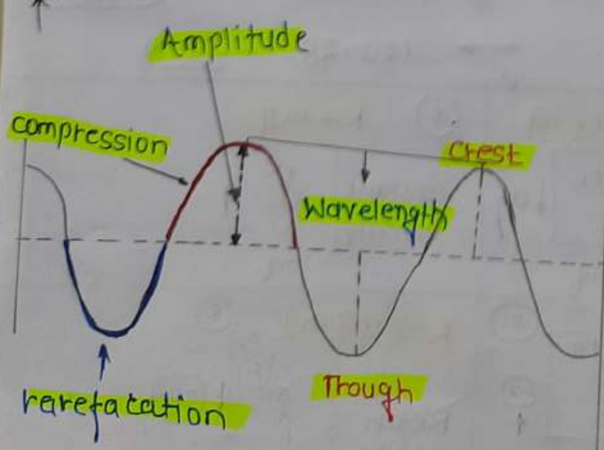
ⓑ sound → mechanical wave

* electromagnetic wave
  ↓
  do not require any medium.

* compression region → high pressure
  rarefaction region → low pressure

* particle → density
  └→ more↑ → medium given more pressure.

*



Amplitude:

→ Size of vibration

→ determines the loudness of sound

- soft sound (Amp.↓)
- louder sound large (Amp).

Frequency:

- osciallate b/w max. value & min. value.
- SI unite – Hz
- Frequecy high → higher pitch → Shrill voice → Ex: woman

  lower frequency → lower pitch → Ex: tiger

Loudness ↑ = Amplitude ↑

$L \propto A^2$

* speed = wavelength × frequency

c – speed

ν – Frequency

λ – wavelength

$c = \nu \times \lambda$

$\boxed{\nu = \frac{c}{\lambda}}$

$\text{Freq.} \propto \frac{1}{\text{wavelength}}$ $\boxed{\nu = \frac{1}{T}}$

* **larynx**: Sound produced in the voice box.

* Range of Hearing.

Wavelength: max → min | Frequency: min → max

- Radio waves
- microwave
- Infrared
- visible light
- Ultraviolet
- x-rays
- Gamma rays

* A sound wave has a freque. of 2KHz and wave length 35cm. How long will it take to travel 1.5 Km?

→

$\nu = 2KHz = 2000\,Hz$

$\lambda = 35\,cm = 0.35\,m$

$t = ?$

$d = 1.5\,km = 1500\,m$

$\text{Speed}(c) = \nu\lambda$

$= 2000 \times 0.35$

$= 700\,m/s$

$\text{time} = \dfrac{\text{Distance}}{\text{Speed}}$

$t = \dfrac{15\cancel{0}0\,\cancel{m}}{70\cancel{0}\,\cancel{m}/s}$

$\boxed{t = 2.1\,s}$

Q. How much time will taken by a sound wave having a frequency of 4KHz & wavelength of 40cm to travel 4 km?

→

$t = ?$

$\nu = 4KHz = 4 \times 1000 = 4000\,Hz$

$\lambda = 40\,cm = \dfrac{4\cancel{0}}{10\cancel{0}} = 0.4\,m$

$d = 4\,km = 4000\,m$

$\text{speed} = \lambda\nu$

$= 0.4 \times 4000$

$c = 1600$

$t = \dfrac{40\cancel{00}}{16\cancel{00}}$

$\boxed{t = 2.5\,s}$

Q. the wavelength of sound wave is 40cm & the time period $T = 0.0005\,s$ how many seconds will it take to travel 4 km?

→

1. ~~5s~~
2. 10s
3. 8s
4. 2s

$\lambda = 40\,cm = 0.4\,m$

$T = 0.0005\,s$

$t = ?$

$d = 4000\,m$

$\boxed{c = \nu\lambda}$

$\text{Frequency}\,(\nu) = \dfrac{1}{T}$

$\nu = \dfrac{1\cancel{0000}}{0.0005}$

$\nu = \dfrac{\overset{2}{\cancel{10}}000}{\cancel{5}}$

$\boxed{\nu = 2000}$

$c = 2000 \times 0.4$

$= 800\,m/s$

$t = \dfrac{40\cancel{00}}{8\cancel{00}}$

$\boxed{t = 5\,s}$

Speed of sound

solid > liquid > gas

Transvere wave	Longitudinal wave
• Trough & creasts • travels only in solid • perpendicular to the direction of propogation. • Ex: Light • E.M waves	• refraction, compression • travel - all state of matter • particles - parallel ↓ Ex: sound - vaccum × ↓ machanical wave

* speed of sound in different media at 25°C.

state	substance	speed m/s.
solids	Aluminium	6420
	Nickel	6040
	steel	5960
	Iron	5950
	Brass	4700
	Glass	3980
Liquid	Water (sea)	1531
	water (distil.)	1498
	Ethanol	1207
	methanol	1103
Gases	Hydrogen	1284
	Helium	965
	Air	346
	oxygen	316
	sulphur dioxide	213.

ECHO

* time interval b/w original & reflected sound - 0.1 s

Q. A person clapped his hands near a cliff and heard the echo after 2s. What is the distance of the cliff from the person if the speed of the sound (v) = 346 m/s

→

speed of sound = 346.

$d = s \times t$

$d = 346 \times 2$

$d = 692$

echo (min dist.) $= \frac{692}{2}$

$\boxed{= 346 m}$

Q. Eco is heard in 3s. what is the distance of the reflecting surface from the source given that the speed of sound is 342 m/s

→ $t = 3s$

$s = 342$ m/s

$D = ?$

$d = 342 \times 3$

$d = 1026$ m

echo $= \frac{1026}{2}$

$= 513$

* Range of Hearing

- audible range - 20Hz to 20KHz (2000Hz)

- children (stage)
- dogs

hear 25 KHz

* infrasonic range → bleow 20Hz
  - whales
  - elephants

* ultrasonic range → above 20KHz
  - dolphins
  - bats
  - porpoise

Inaudible

earthequek - 5Hz
- cat
- dog
- rahino

sonar : → 2d = v × t (speed of sound) (time)

↓

use : ultrasonic waves to measure the distance. (underwater)

* microphone →

sound wave (to) electrical wave.

~~velocity~~ unit of aircarft speed

↓

knot

ultraviolet light

- electromagnetic wave
- travel through vaccum.

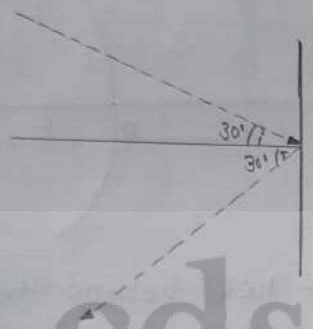
* Lightining
  - speed of sound (less)
  - speed of light (faster)

* ear
  - outer
    - pinna → collect sound → pass through Auditory canal.
  - middle
    - Hammer
    - Anvil
    - stirrup
  - Inner
    - cochlea

Light

* diffraction : light has a tendency to bend.

* law of Reflection of light

- 

- $\angle i = \angle r$
- All lie in the same plane

Plane mirror

- Image formed : virtual & erect
- Image formed always equal in size.
- The Images are laterally inverted.

- Eyes - Real & inverted.
- A non-spherical shining spoon can generally be considered plane mirror. [CDS-2021]

* Number of Image = odd no.

$$n = \frac{360}{\theta} - 1 \left[\frac{360}{\theta} \text{ is even } (-1) \right]$$

$$n = \frac{360}{\theta} \quad \left[\frac{360}{\theta} \text{ is odd} \right]$$

Q. no of image formed by the comb. of two mirror is 9 angle?

1. 45°
2. 44° $\frac{360}{n} = 9$
3. 54°
4. 40° $n = \frac{360}{9}$ 40°

Q. two mirror inclined at some angle θ An object is placed b/w & there ® 5 Images formed θ?

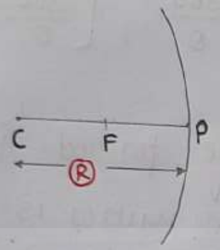
1. 64°
2. 63° $\frac{360}{\theta} = 5$
3. 75°
4. None.

$\therefore \frac{360}{72} = 5$ — ①

Spherical mirror

Reflecting surface curved inwards

Concave mirror

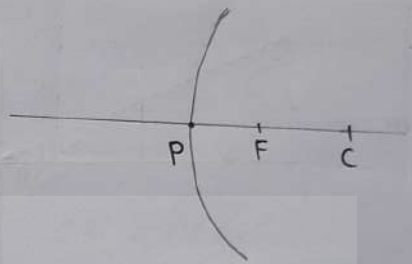


- C → lies in front of it.
- converging mirror
- page catch fire (burn) due to concave mirror.
- uses:

1. torches
2. search lights
3. vehicles headlights
4. shaving mirror - large image of face.
5. dentists
6. produce heat in solar furnaces.

Reflecting surface curved outwards

Convex mirror



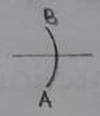
- C → lies behind the mirror
- Diverging mirror.
- uses:

1. Rear view (wing) mirror in vehicles.

- Image - virtual + erect always.

plane mirror → same.

* Aperture - diameter of the reflecting surface.



(AB - Aperture)

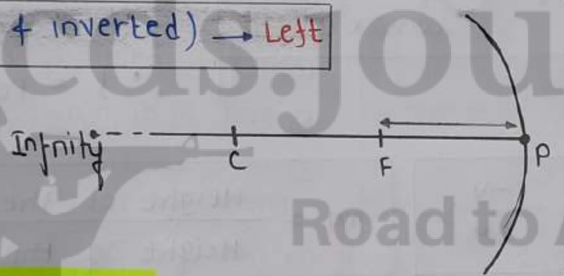
* Radius of curvature (R) * $R = 2f$ (f - focal length)

* centre of curvature is not a part of the mirror.

* Image formation of concave mirror

Position of object	Position of image	Size of image	Nature of image.
At infinity	At Focus (F)	Highly diminished	Real & inverted
Beyond C	Betn F & C	Diminished	Real & inverted
At C	At C	Same sized	Real & inverted
Btn C & F	Beyond C	Enlarged	Real & inverted
At F	At Infinity	Highly Enlarged	Real & inverted
Betn P & F	Behind the mirror.	Enlarged	virtual & erect

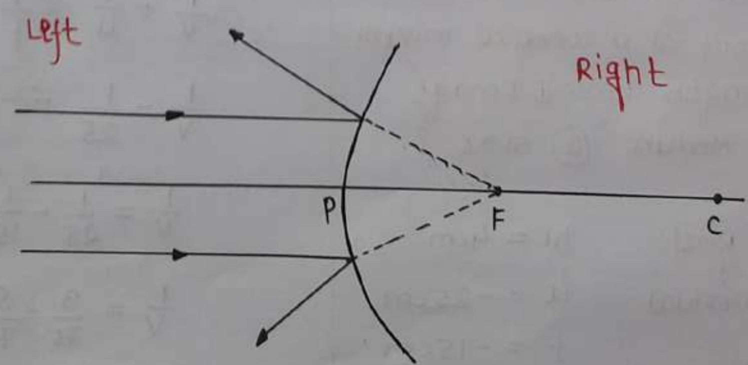
(Real & inverted) → Left



Right :
Image form Right
(virtual & erect)

* Convex mirror.

Position of object	Position of image	Size of image	Nature of image
At inifinity	At Focus (F) behind the mirror.	Higly diminished, point size	virtual & erect
Betn inifinity & Pole(P) of mirror	Betn P & F behind the mirror	Diminished	virtual & erect.



converging

concave mirror | convex lens

Diverging

convex mirror | concave lens

mirror

Left side	Right side
• -ve	• + ve
• Real	• virtual
• u = always ⊖	• u = ⊖
• v = ⊖ (Real)	• v = ⊕ (virtual)
• f = ⊖ concave mirror	• f = ⊕ convex mirror

mirror formula

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

- v = Image distance
- u = object distance
- f = focal length

magnification (m) $= \frac{-v}{u}$

$m = \frac{\text{Height of the image } h_2}{\text{Height of the object } h_1}$

$$m = \frac{h_2}{h_1} = \frac{-v}{u}$$

Positive sign → Image virtual

Negative sign → Image Real

Q. An obj. 4cm in size is placed at 25cm in front of a concave mirror of focal length 15cm ① Image distant ② Nature ③ size ?

→

① 37.5 cm, Real

② 37.5cm, virtual

③ 75cm, real

④ None

$h_1 = 4cm$

$u = -25cm$

$f = -15cm$

$v = ?$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{25} = -\frac{1}{15}$$

$$\frac{1}{v} = \frac{1}{25} - \frac{1}{15}$$

$$\frac{1}{v} = \frac{3}{75} - \frac{5}{75}$$

$$\frac{1}{v} = -\frac{2}{75}$$

$$\boxed{v = -37.5}$$

size

$m = \frac{-v}{u} = \frac{h_2}{h_1}$

$= \frac{\cancel{+}(-37.5)}{\cancel{+}25} = \frac{h_2}{4}$

$= \frac{-37.5}{25} = \frac{h_2}{4}$

$h_2 = \frac{-37.5 \times 4}{25}$

$h_2 = \frac{-150}{25}$

$h_2 = -6cm$

Q. convex mirror used for rear-view on an Automobile has a radius of curvature of 3.00m. If a bus is located at 5.00m from this mirror find the position Nature, size of Image.

$f = 1.5m$
$u = -5m$
$v = ?$

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} - \frac{1}{5} = \frac{1}{1.5}$

$\frac{1}{v} = \frac{10}{15} + \frac{1}{5}$

$\frac{1}{v} = \frac{10+3}{15}$

① $v = 1.15m$

② virtual

③ $m = \frac{-v}{u}$

$m = \frac{\cancel{+}1.15}{\cancel{+}5}$

$m = +0.23$

Q. A concave mirror produce 3 time magnified real image placed at 10cm in front $v = ?$

① 30 cm behind the mirror

② 30cm front of mirror

③ 60 cm " " "

④ None.

→ $m = -3$
$u = -10$
$v = ?$

$m = \frac{-v}{u}$

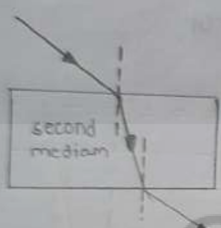
$-3 = \frac{\cancel{+}v}{\cancel{+}10}$

$v = 30$ (-ve)

Sign	
• $u = -ve$	• $u = -ve$
• $v = -$ real	• $v = +ve$ (virtual)
• $m = \frac{-v}{u} = \frac{\cancel{+}(-v)}{\cancel{+}u}$ $m = \frac{-v}{u}$	• $m = \frac{\cancel{+}v}{\cancel{+}u}$ $m = \frac{v}{u}$ $\boxed{m = +ve}$
• magnif(m) = Neg.	• $f = +ve$
• $f = -ve$	
• $v = (-ve)$ fornt of m.	$v = (+)$ behind mirror

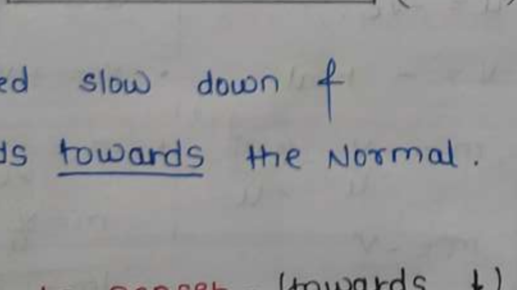
Refraction of Light

Refraction	Reflection
• one medium to second medium	• one medium
• change direction in second medium.	• only direction change.
	• angle same



• solid → Denser	• Air → Rarer
• Liquid → Rarer	• liquid → Denser

① object Rarer to Denser

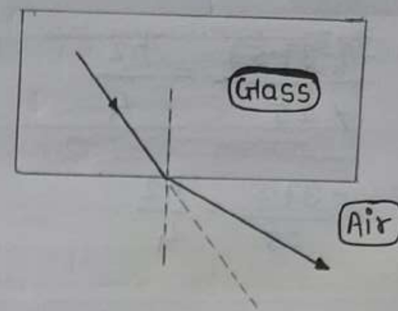


◉ speed slow down & Bends towards the Normal.

◉ Rarer to Denser – (towards ↓)

● Denser to Rarer – away ↑ speed

② Denser to Rarer



◉ speed increase & bend away from Normal.

* coin in water appers slightly raised above its actual position due to refraction of light.

* speed of light

- Air medium > Denser medium
- vaccum > Air

* Snell's law

The ratio of angle of incidence to the sine of angle of refraction is constant.

$$\frac{\sin i}{\sin r} = \text{constant}$$

This constant value is called the refractive index.

$n_{12} = \dfrac{\text{speed of light in medium 2}}{\text{speed of light in medium 1}}$

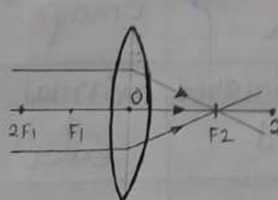
Refractive index

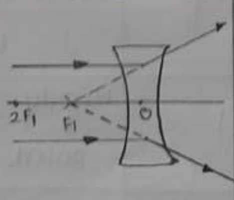
- Air
- Ice
- Water
- Alcohol
- kerosene
- Benzene
- crown Glass
- canada Balsam
- Rock salt
- Dense Flint Glass
- Sapphire
- Diamond

increase ↓

Lenses

- convex lens — converging lens
- concave lens — diverging lens





- lens has two principle Foci → F_1 & F_2
- center point of lens is optical centre → 'O'
- ray of light through the optical centre of lens pass without suffering any deviation.

- **Aperture**: effective diameter of the circular outline of a spherical lens is called Aperture.
  - ~~aperture less than radius of curvature.~~

* Lens

Left ⊖	Right ⊕
• virtual	• Real
• u = ⊖ve always → ✓
• f = (-ve) concave	• f = +ve convex
• v = ⊖ virtual	• v = ⊕ real

mirror	lens
Image → Left ↓ Real + inverted; Right ↓ virtual + erect	Image → Left ↓ virtual; Right ↓ real
$u = (-ve)$ always	$u = -ve$ always
$f = (-ve)$ (concave) $= (+ve)$ (convex)	$f = (+ve)$ convex lens $(-ve)$ concave lens
$v = (+ve)$ - virtual $v = (-ve)$ - real	$v = (+)$ real $v = (-)$ virtual
$m = \frac{-v}{u}$	$m = \frac{v}{u}$
$\boxed{\frac{1}{v} + \frac{1}{u} = \frac{1}{f}}$	$\boxed{\frac{1}{v} - \frac{1}{u} = \frac{1}{f}}$
• If, $v = (+) \longrightarrow m = +ve$ (virtual) • If, $v = (-) \longrightarrow m = (-ve)$ (Real)	If, $v = (+) \longrightarrow m = -ve$ (Real) If, $v = (-) \longrightarrow m = +ve$ (virtual)

Cancave lens

Position of object	Position of Image	Relative size of Image	Nature of Image
At Infinity	At Focus (F_1)	Highly diminished point sized	virtual + erect
Betn Infinity + optical centre O of lens	Betn focus F_1 + optical centre O	Diminished	virtual + erect.

position of object	Position of image	Relative size of image	Nature of image.
At Infinity	At Focus (F_2)	Highly diminished	Real + inverted
Beyond $2F_1$	Betn F_2 + $2F_2$	Diminished	Real + inverted
At $2F_1$	At $2F_2$	sime size	Real + inverted
Betn F_1 + $2F_1$	Beyond $2F_2$	Enlarged	Real + inverted
At Focus F_1	At infinity	highly Enlarged	Real + inverted
Betn focus F_1 + optical centre O	on the same side of lens as object	Enlarged	virtual + erect.

Q. A concave lens has focal length of 15cm. At distance showd the object from the lens be placed so that it forms an image at 10 cm from the lens? magn. of lens?

→

$f = -15cm$

$V = -10cm$

$u = ?$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{u} = \frac{1}{v} - \frac{1}{f}$$

$$\frac{1}{u} = \frac{1}{-10} + \frac{1}{15}$$

$$\frac{1}{u} = \frac{-3}{30} + \frac{2}{30}$$

$$\frac{1}{u} = \frac{-1}{30}$$

$$u = -30cm$$

mag. →

$$m = \frac{v}{u}$$

$$m = \frac{\cancel{-}10}{\cancel{-}30} = (+)0.33$$

Power of lens

↓

the power of lens is defined as the reciprocal of its focal length

$$P = \frac{1}{f}$$

- S.I unit of power – dioptre (D)
- $f = m = P = D$
- $1D = 1m^{-1}$

power

convex lens (+)

Ex: $P = +2\,D$

$f = 0.50m$.

concave lens (−)

Ex – $P = -2.5\,D$

$f = -0.40m$

Electricity

- **charge** → Atom's property of being +ve (or) -ve

  ←+ +→ , +→ ←− ←− −→

- SI unit → coulomb (C)

- $1C = 6.25 \times 10^{18}$ electrons

- current = $\dfrac{\text{charge}}{\text{time}}$

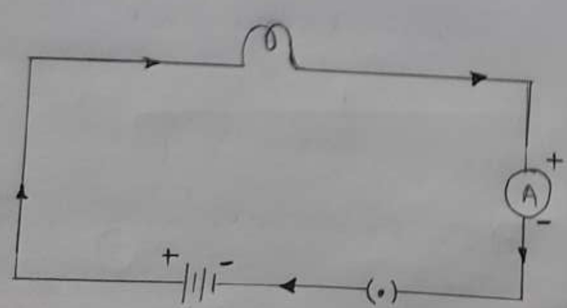
$$\boxed{I = \frac{Q}{t}}$$

- SI unit of current – Ampere (A)

  - $1mA = 10^{-3}A$ (milli.)
  - $1\mu A = 10^{-6}A$ (micro)

* **Ammeter** → measure electric current in circuit

Always connected in series.



electric current flow (+) to (−)

Q. current of 0.5A is drawn by a filament of an electric bulb for 10 minutes. find the Amount of electric charge that flow through the circuit.

→

$I = 0.5A$

$t = 10 min. = 10 \times 60 = 600S$

$Q = It$

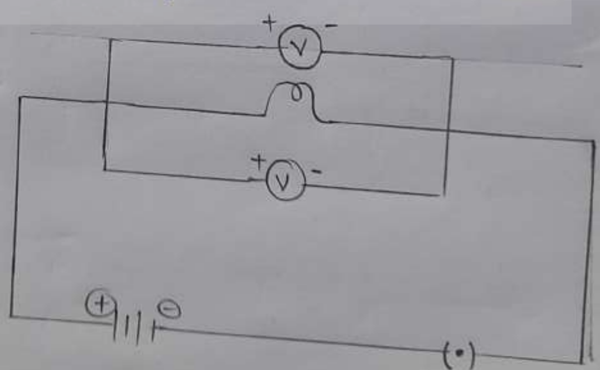
$Q = 0.5 \times 600$

$Q = 300C$

\# **Voltmeter** ← measured potensial diff. (V).

- ~~SI Unit~~ –

- Alway connected in parallel.



Ammeter	voltmeter
• Always connected in series	• Always connected in parallel

potential difference :

→ electrons moves only if there is a difference of electric pressure.

- $V = \frac{\text{Work done}}{\text{charge}}$

$$\boxed{V = \frac{W}{Q}}$$

- SI unit of V — Volt (V)

Q. How much work is done in moving a charge of 2C across two points having a potensial difference 12V?

→ $W = ?$

$V = 12V$

$Q = 2C$

$V = \frac{W}{Q}$

$W = VQ$

$W = 12 \times 2$

$= 24J$

$$\boxed{1V = 1JC^{-1}}$$

Ohm's law.

- V - potensial difference
- I - current

$V \propto I$

$\frac{V}{I} = \text{constant}$

$= R$

$$\boxed{V = IR}$$

→ Resistance

property of a conductor to resist ~~charge~~ flow of charge.

- SI unit = Ω (ohm)

$$\boxed{R = \frac{V}{I}}$$

- If the resistance is doubled the current gets halved.

Resistances

- directly proportional to length
- inverly proportional to area
- Nature of the material.

$$\boxed{R = \rho \frac{l}{A}}$$

(ρ - Resistivity)

* rheostart : used
↓
to charge the resistance in the circuit.

Resistance

- higher Resistance = poor conductor of electricity
- low Resistance = good conductor.

① $R \propto l$ (length)

② $R \propto \frac{1}{A}$

$R \propto \frac{l}{A}$

$\boxed{R = \rho \frac{l}{A}}$

Resistivity → SI unit = Ω

Insulator (max.) > Alloys > conductor (min)

* which is correct increasing order of resistivity.

① Tungten < Nichrome < Glass.

(#) resistivity in decreasing order.

Insulator
1] paper dry
2] Diamond
3] Ebonite
4] Hard rubber
5] Glass

Alloys
6] Nichrome [alloy - Ni, Cr, Mn, Fe]
7] Manganin [alloy of Cu + Ni]
8] constantan [alloy of Cu + Ni]

~~Insulator~~
conductor
9] mangnese
10] Mercury
11] chromium
12] iron
13] Nickel
14] Tungsten
15] Aluminum
16] copper 17] silver

Q. A wire of given material having length l, Area A has resis. of 4 ohm what would be the res. of another wire of same material having length $\frac{l}{2}$ and area 2A.

① 2Ω
② 3Ω
③ 5Ω
④ 1Ω

$R = \rho \frac{l}{A}$ —①

$R' = \rho \frac{\frac{l}{2}}{2A}$

$R' = \rho \frac{l}{2 \times 2A} = \frac{1}{4}\left(\rho \frac{l}{A}\right)$

$R' = \frac{1}{4} \times 4$

$\boxed{R' = 1\Omega}$

Q. A wire of length l & Resis. (R) is stretched so that the length is doubled and Area halved.

→ ① How will Resistance change ② Resistivity

① $\frac{1}{4}$th

② $\frac{1}{2}$nd

③ 4 times

④ 2 times

$$R = \rho \frac{l}{A}$$

$$R' = \frac{\rho\, 2l}{\frac{A}{2}}$$

$$R' = 4 \frac{\rho l}{A}$$

$$\boxed{R' = 4} \text{ — ①}$$

② Resistivity does not depend on

(i) area or length

* depend on

(i) material

(ii) temp.

(iii) Nature.

Q. A cylindrical wire of length l & radius (r) has a res. (R). The resis of another wire of same material but having four time its length & half its radius will be.

① 16R

② 1/16R

③ 8R

④ R

$$R = \rho \frac{l}{A} \text{ — ①}$$

Aarea $(A) = \pi r^2$

$$R = \rho \frac{l}{\pi r^2}$$

$$R' = \frac{\rho 4l}{\frac{\pi r^2}{4}}$$

$$R' = 16 \frac{\rho l}{\pi r^2}$$

$$R' = 16R$$

(#) A fuse wire should have

- low melting point
- high risistance.
- Fuse used for domastic purpose rated
  - 1A
  - 2A
  - 3A
  - 5A
  - 10A

Resistor

Resistor in series is equal to the (V) across the individual resistors.

$$\boxed{V = V_1 + V_2 + V_3}$$

$$V = IR$$

$$\boxed{R = R_1 + R_2 + R_3}$$

parallel is equal to the sum of the reciprocal of the individu. resistance.

$$\boxed{\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

Heating effect of electric cur.

ususful

- electric laundry iron
- electric toaster
- electric oven
- electric kettle
- also used to produce light, as in an electric bulb.

↓

tungsten

used for making bulb filaments.

Electric power

- SI unit of - Watt (W)

$$\boxed{P = VI}$$

$$\boxed{P = I^2R = V^2/R}$$

$1W = 1V \times 1A = \underline{1VA}$

$1KWh = 1000\,watt$

$(1KWh = 3.6 \times 10^6\, Jule)$

$Heat\,(H) = VIt$

$$\underline{H = I^2RT}$$

Q. An electric bulb is connected to a 220 V generator. the current is 0.50 A. What is the power of the bulb.

→ $V = 220V$

$I = 0.50A$

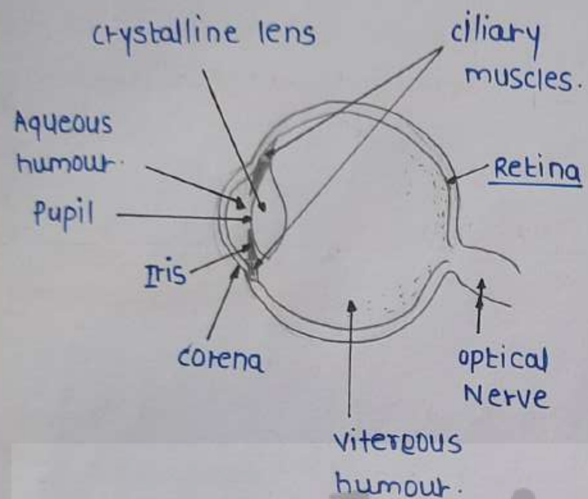
$P = ?$

$P = VI$

$= 220 \times 0.50$

$\underline{P = 110W}$

⑥ Human Eye & colour.



rod cells
→ light & dark

cone cell
→ distingtue of colour.

Iris - control size of pupil.

pupil - Regulates the amount of light.

Retina - Its lens system forms an images on a light-sensitive screen called retina.

↓

real & inverted Image.

corena - Light enter the eye through corena.

eyeball - diameter about 2·3 cm.

corena - we can donate.

Q] why do star twinkle?

→ Refraction. ↓ (mugal)

due to atmospheric refraction.

→ eye lens is composed of a fibrous, jelly-like material

→ ability of the eye lens to adjust its focal length is called accommodation.

→ see - 25cm (good for eye).

→ crystalline lens of people at hold age becomes milky and cloudy this condition is called cataract.

→ A human beings has a horizontal field of view of about 150° with one eye.

→ two eye - about 180

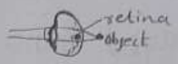
→ The vision become blurred due to the refractive defects of the eye.

Myopia

- Near sightedness. ✓
- Near by object clearly.
- the image of a distant object is formed in front of the retina.
- we can use cancave lens for Image at Retina.
- also known Diverging lens
- Focal length is small.

Hypermetropia

- Far - sightedness.
- see distant object clearly.
- 25+cm.
- the Focal length of the object eye lens is too long.



- eyeball has become to small.
- defect can be corrected using a convex lens.
- converging lens.

presbyopia.

- the power of accommodation of the eye usually decrease with aging.
- difficult to see Nearby object comfortably.
- distinctly without corrective eye-glasses
- due to gradual wekening of the cliary muscles.
- require bi-Focal lense.

↓

both cancave & convex lens.

- upper Portion
  ↓
  concave lens
  ↓
  lower part
  ↓
  convex len.

IMP.

- one pair of eyes gives vision to TWO corneal Blind people.

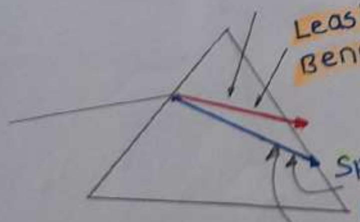
- can donate eye. (died).
  - diabetic
  - hypertension
  - asthama

- can not donate eye (died).
  - AIDS
  - Hepatitis B or C.
  - rabies
  - acute leukaemia
  - tetanus
  - cholera
  - meaningitis or encephalitis.

prism

(#) speed of Red light max.

Least Bend.



Speed of Violet light minium.

Max. bend.

- violet (speed min.)
- indigo
- Blue
- Green
- yellow
- orange
- Red.

- Splitting of light into its component coloures is called dispersion.

What causes of Rainbow?

→ Dispergin of light.

Rainbow

→ always formed in a direction opposite to that sun.

IMP

Advance Sunrise & Delayed sunset.

→ the sun is visible to us about 2 mintutes [redacted] before the actual sunrise, and about 2 minutes after the actual sunset because

↓

atmospheric refraction

Scattering of light.

→ The blue colour of the sky.
colour of water in deep sea.

→ blue colour is max. scatter.

→ Red colour is less scatter.

→ blue has ~~least~~ least wavelength.

→ Red → ~~more~~ more wavelength.

→ Red light has wavelength about 1.8 times greater than blue light.

→ sun red because of scattering

→ [redacted].

1] the human eye can focus on objects at different distance by adjesting the focal length of the eye lens. This is due to

→ Accommodation